Spatio-Temporal Momentum — Backtest Report

Model: STM (TFT)
Period: 2026-09-25 to 2026-09-29 (3 sessions)
Universe (22): VZ, GOOGL, AMZN, HD, PG, KO, CVX, XOM, BAC, JPM, JNJ, PFE, HON, CAT, MSFT, AAPL, NEM, LIN, PLD, SPG, DUK, NEE

Layout follows Tan, Roberts & Zohren (2023), arXiv:2302.10175.

Notes:
- DMN (reference model of the paper) is not implemented in this project and is not reported.
- TSMOM / CSMOM use the longest configured return horizon (21 days); the paper uses 252 days.
- Rescaled results target 15% annualised volatility with an ex-ante 20-day EWM estimate.
- Rolling correlations use a 20-session window (paper: 1 year).
- SHAP values: Expected Gradients (shap.GradientExplainer formulation); the paper uses Deep SHAP.

Table 1: Performance Metrics for Strategies – Raw Signal Outputs

	E[Return]	Vol.	Downside Deviation	MDD	Sharpe	Sortino	Calmar	Hit Rate	Ave. P / Ave. L
Long Only	-0.880	0.018	0.057	0.010	-50.230	-15.371	-84.284	0.000	–
TSMOM	0.449	0.054	0.017	0.002	8.290	26.388	241.849	0.667	1.940
MACD	0.125	0.011	0.003	0.000	11.085	49.427	453.004	0.667	3.196
CSMOM	-0.171	0.007	0.012	0.002	-25.538	-14.156	-84.050	0.000	–
STM (TFT)	0.104	0.035	0.012	0.002	2.944	8.618	60.883	0.333	3.449

Table 2: Performance Metrics for Strategies – Rescaled to Target Volatility

	E[Return]	Vol.	Downside Deviation	MDD	Sharpe	Sortino	Calmar	Hit Rate	Ave. P / Ave. L
Long Only	–	–	–	–	–	–	–	–	–
TSMOM	–	–	–	–	–	–	–	–	–
MACD	–	–	–	–	–	–	–	–	–
CSMOM	–	–	–	–	–	–	–	–	–
STM (TFT)	–	–	–	–	–	–	–	–	–

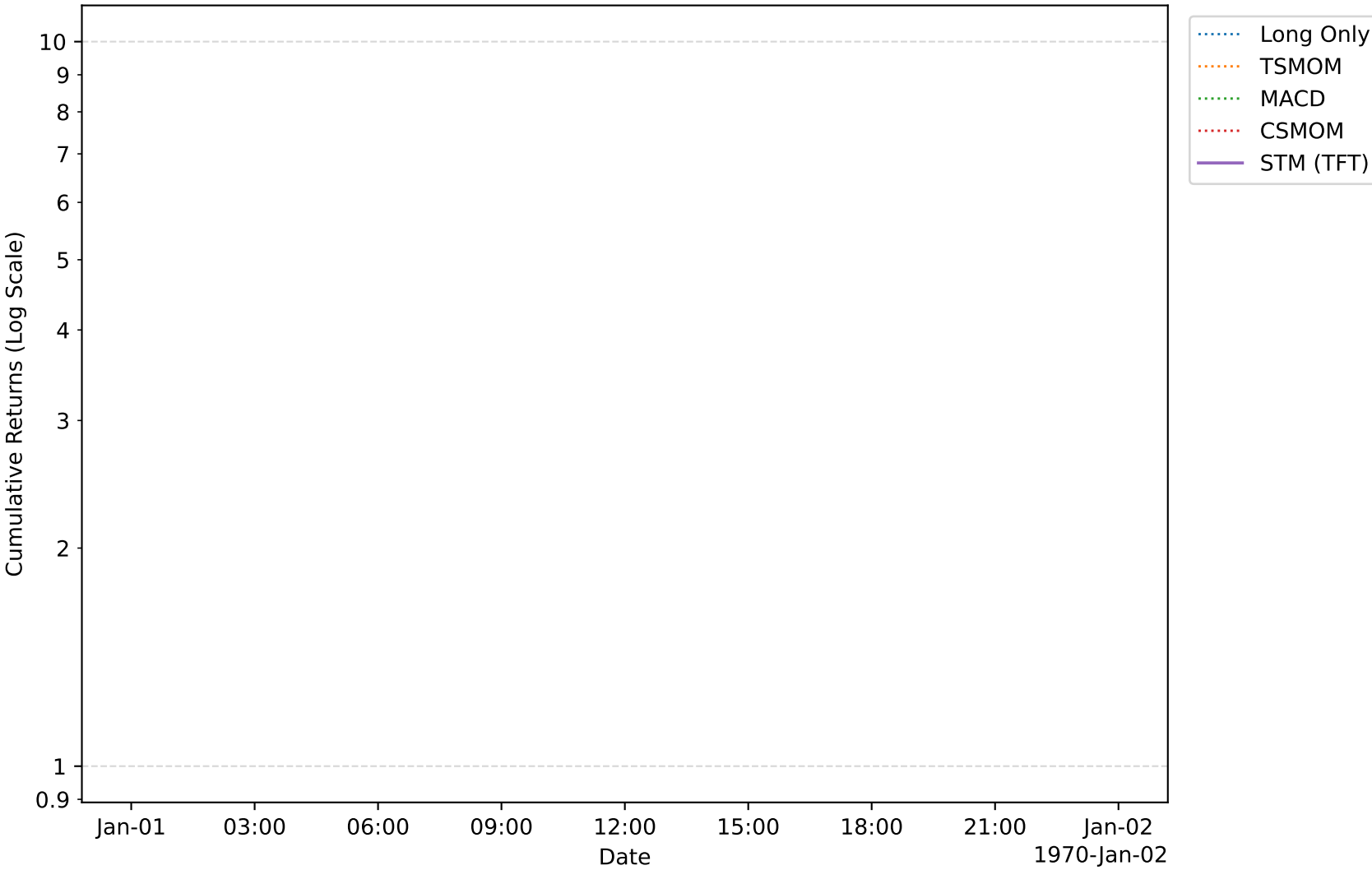

Figure 1: Cumulative Returns - Rescaled to Target Volatility
Cumulative Returns (Log Scale)
10
9
8
7
6
5
4
3
2
1
0.9
Jan-01
03:00
06:00
09:00
12:00
15:00
18:00
21:00
Jan-02
Date
1970-Jan-02
Long Only
TSMOM
MACD
CSMOM
STM (TFT)

Figure 3: (Left) Correlation of Strategies (Right) Rolling Correlations

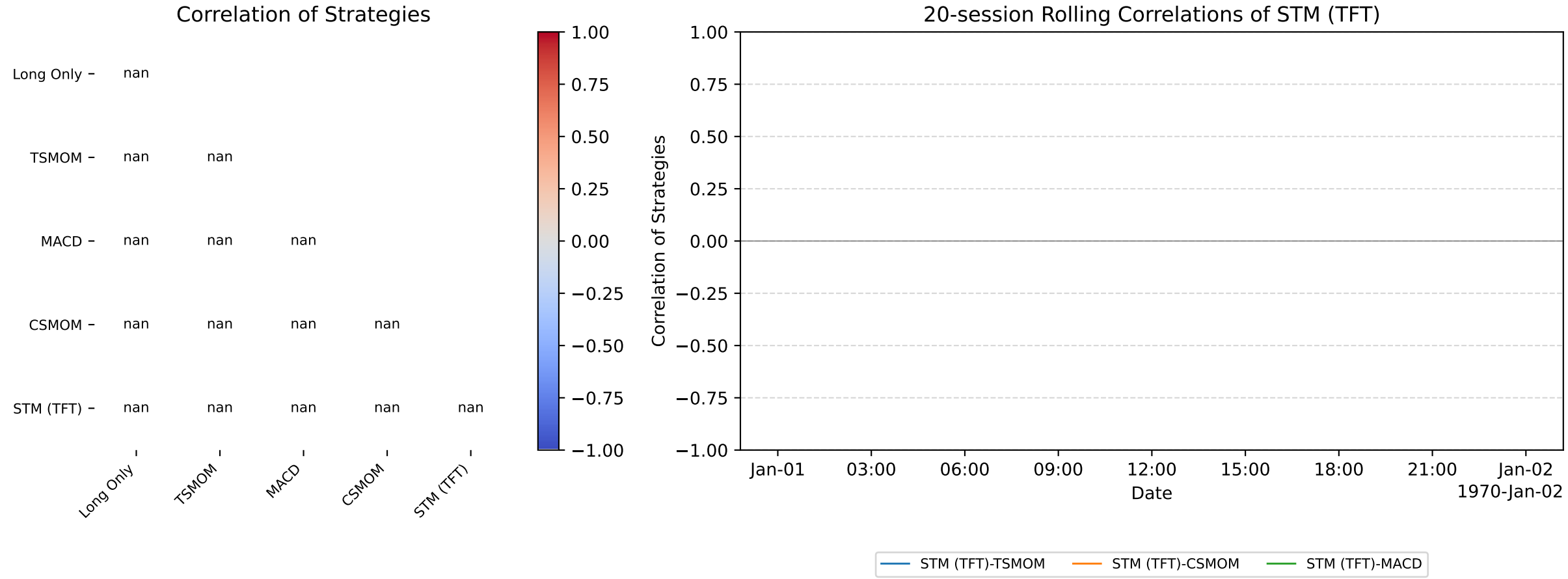
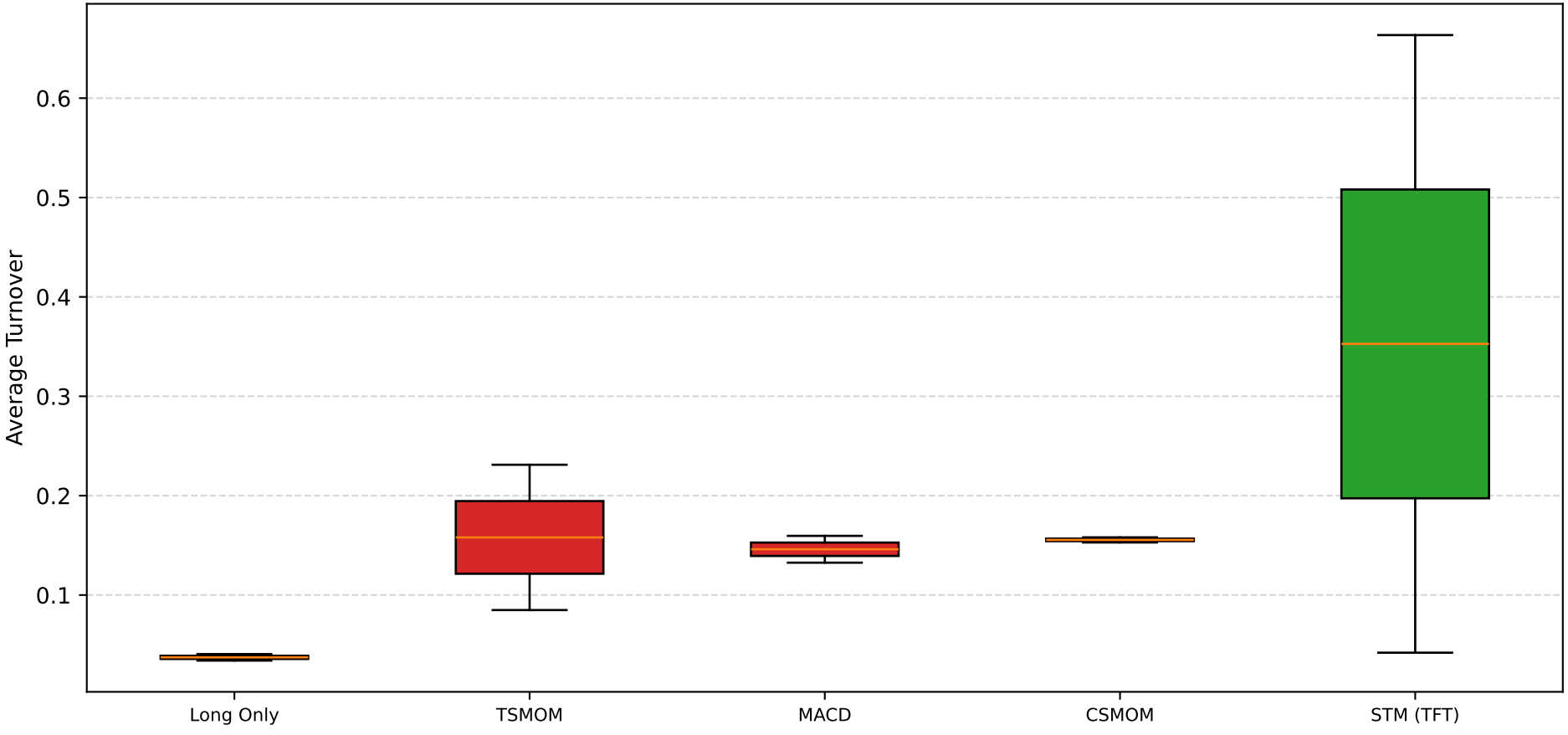
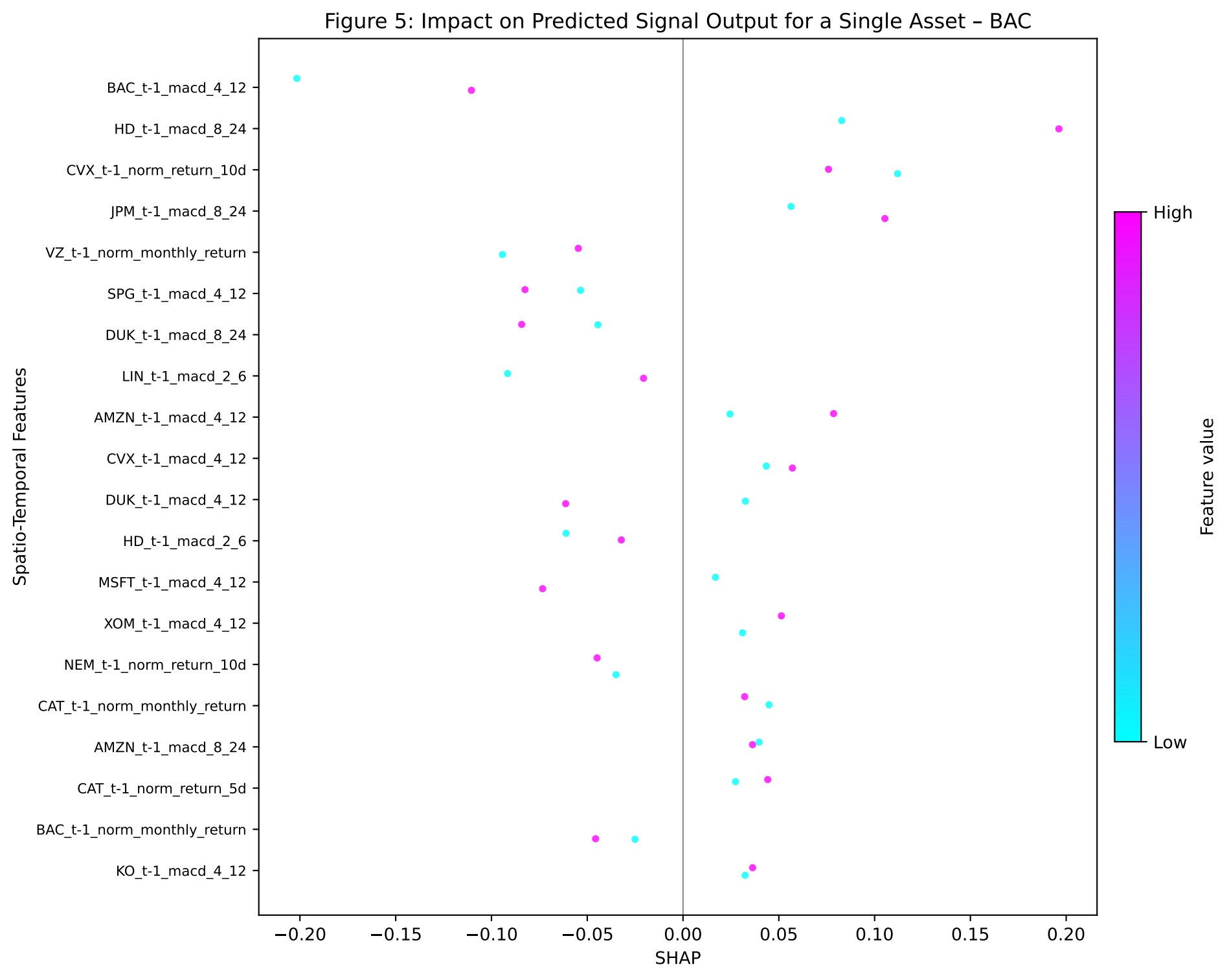
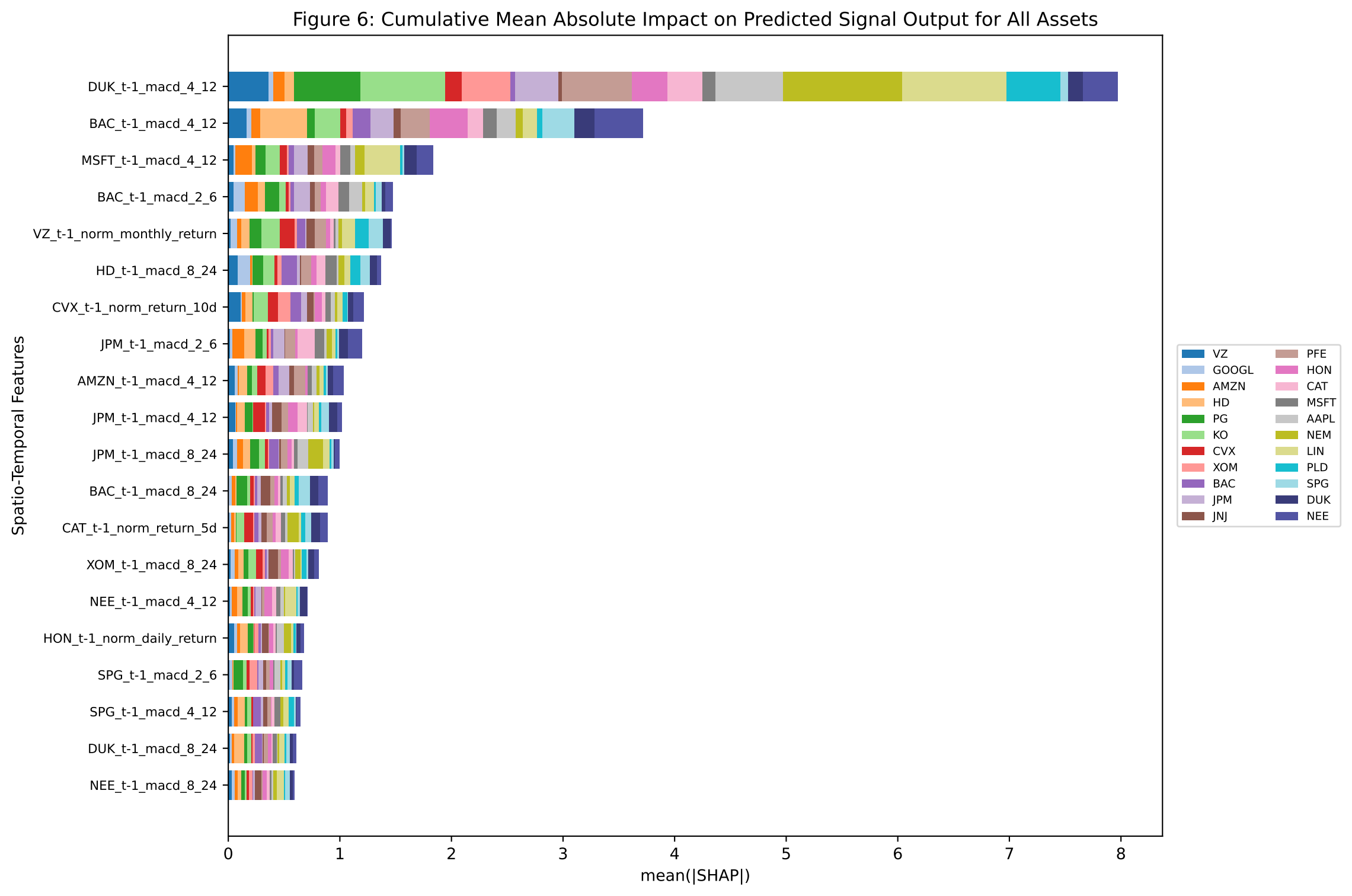


Table 6: Impact of Transactions Costs on Sharpe Ratio – Rescaled to Target Volatility (columns: transaction cost in basis points)

	0	0.5	1	2	3	4	5	10
Long Only	–	–	–	–	–	–	–	–
TSMOM	–	–	–	–	–	–	–	–
MACD	–	–	–	–	–	–	–	–
CSMOM	–	–	–	–	–	–	–	–
STM (TFT)	–	–	–	–	–	–	–	–


Figure 4: Average Turnover of Strategies
Average Turnover
0.6
0.5
0.4
0.3
0.2
0.1
Long Only
TSMOM
MACD
CSMOM
STM (TFT)


Figure 5: Impact on Predicted Signal Output for a Single Asset – BAC
BAC_t-1_macd_4_12
HD_t-1_macd_8_24
CVX_t-1_norm_return_10d
JPM_t-1_macd_8_24
VZ_t-1_norm_monthly_return
SPG_t-1_macd_4_12
DUK_t-1_macd_8_24
LIN_t-1_macd_2_6
AMZN_t-1_macd_4_12
CVX_t-1_macd_4_12
DUK_t-1_macd_4_12
HD_t-1_macd_2_6
MSFT_t-1_macd_4_12
XOM_t-1_macd_4_12
NEM_t-1_norm_return_10d
CAT_t-1_norm_monthly_return
AMZN_t-1_macd_8_24
CAT_t-1_norm_return_5d
BAC_t-1_norm_monthly_return
KO_t-1_macd_4_12
Spatio-Temporal Features
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
SHAP
High
Low
Feature value


Figure 6: Cumulative Mean Absolute Impact on Predicted Signal Output for All Assets
DUK_t-1_macd_4_12
BAC_t-1_macd_4_12
MSFT_t-1_macd_4_12
BAC_t-1_macd_2_6
VZ_t-1_norm_monthly_return
HD_t-1_macd_8_24
CVX_t-1_norm_return_10d
JPM_t-1_macd_2_6
AMZN_t-1_macd_4_12
JPM_t-1_macd_4_12
JPM_t-1_macd_8_24
BAC_t-1_macd_8_24
CAT_t-1_norm_return_5d
XOM_t-1_macd_8_24
NEE_t-1_macd_4_12
HON_t-1_norm_daily_return
SPG_t-1_macd_2_6
SPG_t-1_macd_4_12
DUK_t-1_macd_8_24
NEE_t-1_macd_8_24
Spatio-Temporal Features
0
1
2
3
4
5
6
7
8
mean(|SHAP|)
VZ
GOOGL
AMZN
HD
PG
KO
CVX
XOM
BAC
JPM
JNJ
PFE
HON
CAT
MSFT
AAPL
NEM
LIN
PLD
SPG
DUK
NEE